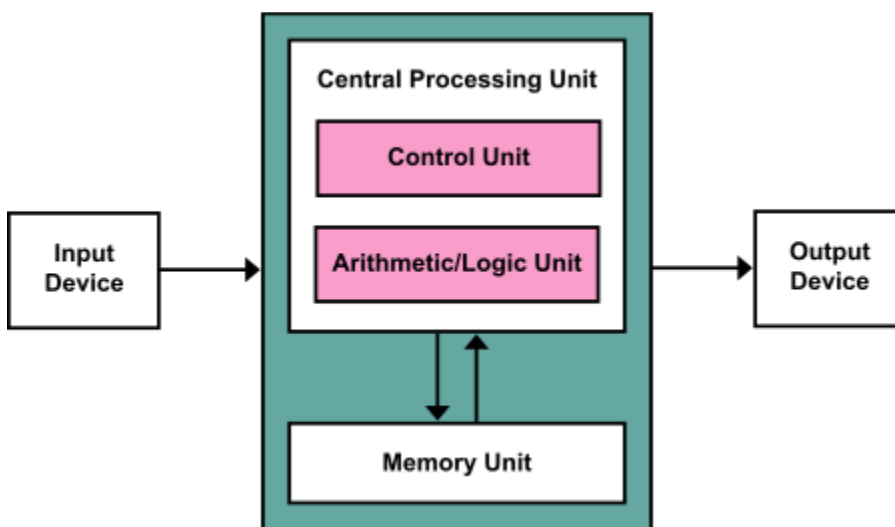


Von Neumann architecture:

Von-Neumann proposed his computer architecture design in 1945 which was later known as Von-Neumann Architecture. It consisted of a Control Unit, Arithmetic, and Logical Memory Unit (ALU), Registers and Inputs/Outputs.

In this stored-program concept, programs and data are stored in a separate storage unit called memories and are treated the same.

Block Diagram:



Components of Von-Neumann Model:

- Central Processing Unit
- Buses
- Memory Unit

Central Processing Unit:

The part of the Computer that performs the bulk of data processing operations is called the Central Processing Unit and is referred to as the CPU.

The Central Processing Unit can also be defined as an electric circuit responsible for executing the instructions of a computer program.

The CPU performs a variety of functions dictated by the type of instructions that are incorporated in the computer.

It has following major components:

1. Control Unit(CU)
2. Arithmetic and Logic Unit(ALU)
3. variety of Registers

Control Unit –

The Control Unit of a computer system controls the operations of components like ALU, memory and input/output devices.

The Control Unit consists of a program counter that contains the address of the instructions to be fetched and an instruction register into which instructions are fetched from memory for execution

Arithmetic and Logic Unit (ALU) –

The arithmetic logic unit is that part of the CPU that handles all the calculations the CPU may need, e.g. Addition, Subtraction, Comparisons. It performs Logical Operations, Bit Shifting Operations, and Arithmetic operations.

Registers –

Registers refer to high-speed storage areas in the CPU. The data processed by the CPU are fetched from the registers. There are different types of registers used in architecture :-

1. **Accumulator:** Stores the results of calculations made by ALU. It holds the intermediate of arithmetic and logical operations. It acts as a temporary storage location or device.
2. **Program Counter (PC):** Keeps track of the memory location of the next instructions to be dealt with. The PC then passes this next address to the Memory Address Register (MAR).
3. **Memory Address Register (MAR):** It stores the memory locations of instructions that need to be fetched from memory or stored in memory.
4. **Memory Data Register (MDR):** It stores instructions fetched from memory or any data that is to be transferred to, and stored in, memory.
5. **Current Instruction Register (CIR):** It stores the most recently fetched instructions while it is waiting to be coded and executed.
6. **Instruction Buffer Register (IBR):** The instruction that is not to be executed immediately is placed in the instruction buffer register IBR.

Buses –

Data is transmitted from one part of a computer to another, connecting all major internal components to the CPU and memory, by the means of Buses. Types:

Data Bus: It carries data among the memory unit, the I/O devices, and the processor.

Address Bus: It carries the address of data (not the actual data) between memory and processor.

Control Bus: It carries control commands from the CPU (and status signals from other devices) in order to control and coordinate all the activities within the computer.

Input/Output Devices –

Program or data is read into main memory from the input device or secondary storage under the control of CPU input instruction. Output devices are used to output information from a computer. If some results are evaluated by the computer and it is stored in the computer, then with the help of output devices, we can present them to the user.

Memory Unit

A memory unit is a collection of storage cells together with associated circuits needed to transfer information in and out of the storage. The memory stores binary information in groups of bits called words. The internal structure of a memory unit is specified by the number of words it contains and the number of bits in each word.

Two major types of memories are used in computer systems:

- RAM (Random Access Memory)
- ROM (Read-Only Memory)

Von Neumann bottleneck –

Whatever we do to enhance performance, we cannot get away from the fact that instructions can only be done one at a time and can only be carried out sequentially. Both of these factors hold back the competence of the CPU. This is commonly referred to as the 'Von Neumann bottleneck'. We can provide a Von Neumann processor with more cache, more RAM, or faster components but if original gains are to be made in CPU performance then an influential inspection needs to take place of CPU configuration.